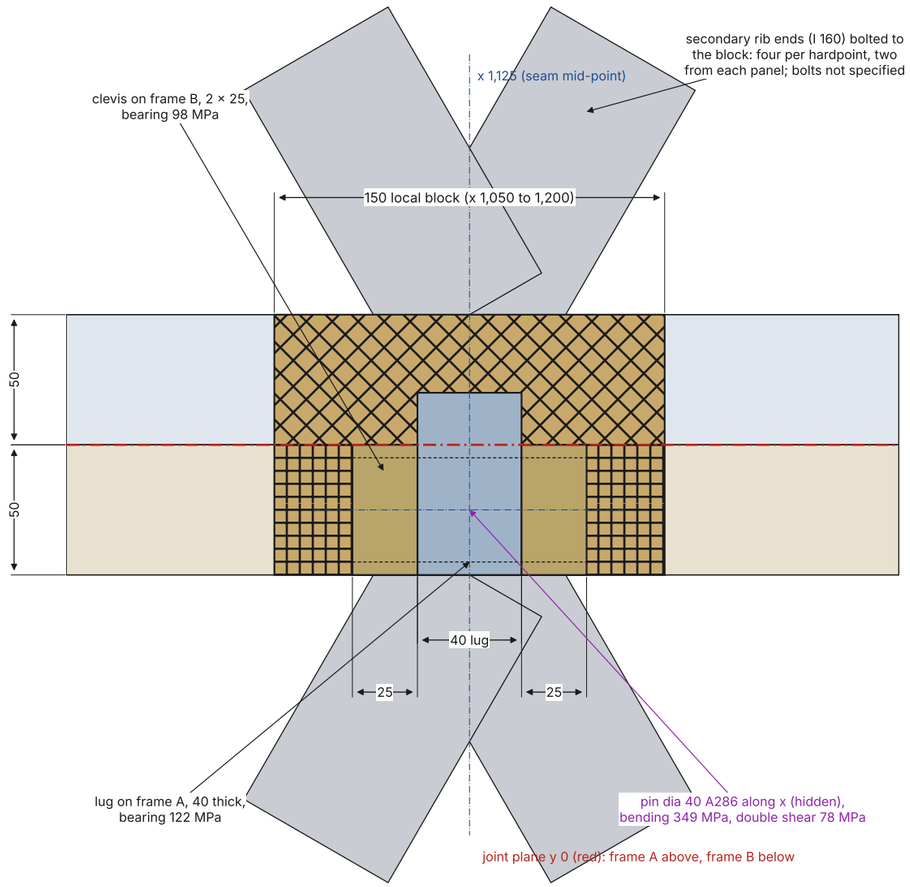
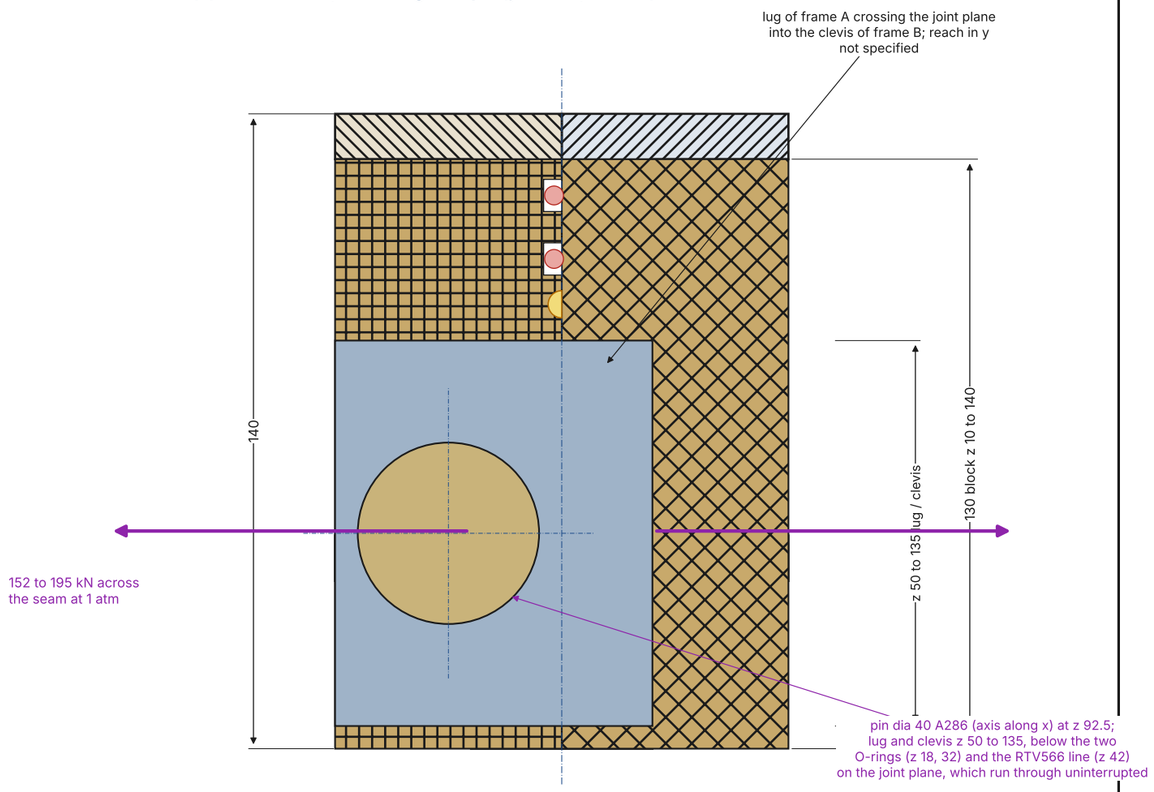
(a) Plan at the seam mid-point, viewed from the cabin (z 140 face): x along the seam, y across



(b) Section at x 1,125 through the pin (y across, z down): block z 10 to 140 in both frames



pinned: no moment about the seam (Rev D intent); stiffness at least 1e7 N/mm

Spec section 6
At the mid-point of every seam (90 on the shell), where four secondary ribs land (two from each panel). The lug on frame A (40 thick) sits between a clevis on frame B (2 x 25), overlapping across the joint plane; the pin dia 40 A286 runs along the seam (x) so the joint carries 152 to 195 kN across the seam (10 and 12 Oct runs) and no moment about it. Hand checks at 195 kN: pin bending governs, M = P/2 x (t_clevis/2 + t_lug/4) = 2.19 kN m, 349 MPa on Z = 6,283 mm3 (a dia 30 pin would see 827 MPa, above A286 yield: out); double shear 78 MPa; bearing 122 lug, 98 clevis.
Lug and clevis occupy z 50 to 135 so the three seal lines (z 18, 32, 42) run through above them. Local block 150 long (x) in both frames, z 10 to 140, secondary rib ends bolted to it. Stiffness at least 1e7 N/mm. If the built detail is moment-stiff the secondary rib becomes I 200 (RD-04, OPEN 1).
Seam bolt station 15 (x 1,125) is taken by the pin: the seam has 26 seal-clamp bolts at the other stations (RD-05). PROPOSAL, C3.

PROPOSAL: sized in C3, 16 Oct

Rev D: FEA-sized baseline (9-12 Oct runs). PROPOSAL items are drawn to carry FEA loads and are sized in the 16 Oct Ansys submodels.

Not yet locked in TRUTH until Manos signs the Rev D revision. Pin y position and block y-extent: not specified (C3).

HCSA | Rev D blueprint pack	Rev D 14 Oct 2026 NTS: do not scale. mm
Sheet 07: seam mid hardpoint at x 1,125, plan and section (PROPOSAL)	**DWG RD-07**
Material: Blocks, lug and clevis 6061-T651; pin dia 40 A286	
HT: T651: solution 530 C, water quench, stretch 1.5-3 %, age 160 C 18 h. Never welded, never above 150 C. A286: solution 980 C, age 720 C 16 h	